Extending DreamCatalyst: Adaptive Guidance, Scene-Aware Localization, and Multi-View Consistency for Text-Driven 3D Editing

Arthur Gislon Leonida
Aspiring AI Researcher in Generative Modeling, Diffusion Guidance, 3D Vision, and Medical AI
B.Sc. Control and Automation Engineering, UFSC (Brazil)
arthurleonida@gmail.com

1 Thesis at a Glance

Goal. Extend DreamCatalyst [1] so that a *single guidance stack* delivers strong text-driven 3D edits and clean background preservation across diverse scenes, auto-adapting to scene type via a scalar *edit-strength* signal read directly from the raw noise predictions, with no per-scene re-tuning.

Two-phase evaluation. **Part 1** is a universal 2D guidance/localization/preservation configuration, compared against the DreamCatalyst baseline (Table 1). **Part 2** is an optional 3D voxel-mask cache for multi-view consistency, evaluated *separately* as a module on top of Part 1 (Table 2); this is why the cache does not drive the Part 1 metrics.

Headline results.

- **Part 1: preservation wins at tied editability.** One fixed configuration (no per-scene tuning) across 8 edits on 5 NeRF scenes: **SSIM improved on all 8**, **LPIPS reduced on 7 of 8** (mean −9.2%), and CLIP-I improved on 7 of 8, with visibly cleaner background line patterns, color fidelity, and material textures. Editability is held, not traded: mean $CLIP_{\text{dir}}$ $0.151 \to 0.152$ (largest gains stormtrooper +10.3%, Batman +5.8%).
- **Part 2: multi-view consistency.** The optional 3D voxel cache is metric-neutral in aggregate, but it makes residual cross-view failures *coherent* across cameras (a region is edited in all views or in none, rather than flickering between them).
- **Three transferable findings.** Localization masks must be built from a *pre-guidance* noise snapshot, or they inherit TAG/STG amplification artifacts; STG's effect is scene-type-dependent (motivating a *bump* schedule); and cross-view 3D consistency needs a 3D *memory*, not merely a smoother 2D mask.

Pipeline. (i) COLMAP structure-from-motion → (ii) Nerfstudio *nerfacto* reconstruction → (iii) Delta Denoising Score (DDS) [2] editing with an InstructPix2Pix [3] backbone → (iv) optional SDEdit-based refinement. Our contributions live entirely in stage (iii) and operate *after* classifier-free guidance, *before* the DDS gradient is taken. They are modular components that can each be switched off; with all of them off the editor is exactly DreamCatalyst.

Landscape. This work connects to recent score-distillation methods on identity preservation and editability, including PDS [6], DaCapo [7], RoMaP [10], Piva [8], SSD [9], and UDS [11]. Our contribution is to combine *mask-based localization* with separate masks for the edit and for preservation, *3D-consistent mask aggregation*, *latent-statistic anchoring*, and *edit-strength-adaptive preservation* inside one modular DreamCatalyst extension.

2 Contributions

We organize the contributions into five groups: **G1** guidance amplification (Sec. 2.1), **G2** per-view localization (Sec. 2.2), **G3** cross-view localization (Sec. 2.3), **G4** preservation anchors (Sec. 2.4), **G5** adaptive mechanisms and universal configuration (Sec. 2.5).

2.1 G1 — Guidance Amplification

[A] Adaptive TAG. TAG [4] decomposes the predicted noise $\boldsymbol{\epsilon}$ into radial $\boldsymbol{\epsilon}_{\parallel}$ (along $\mathbf{v} = \mathbf{z}_t/\|\mathbf{z}_t\|$) and tangential $\boldsymbol{\epsilon}_{\perp}$ components: $\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\parallel} + \eta\,\boldsymbol{\epsilon}_{\perp}, \eta > 1$. Fixed η overamplifies low-noise timesteps. We introduce a concave schedule:

$$\eta(\hat{t}) = 1 + (\eta_{\max} - 1)\,\hat{t}^{\,1/e}, \tag{1}$$

with $1/e \approx 0.368$. The decay naturally aligns with DreamCatalyst's own $w_{\text{DDS}}(\hat{t})$ term.

[B] Asymmetric TAG. DDS uses $\Delta\boldsymbol{\epsilon} = \boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}$. We apply TAG only to the target ($\eta_{\text{tgt}} = \eta(\hat{t})$, $\eta_{\text{src}} = 1$), so the source stays the unmodified reconstruction that DDS subtracts. Because TAG is linear in the prediction, the amplified delta is $\Delta\boldsymbol{\epsilon} + (\eta - 1)\,\boldsymbol{\epsilon}_{\text{tgt},\perp}$: the extra term is the tangential part of the *whole* target prediction, not of the delta.

[C] STG with a bump schedule. Following STG [5], a weak UNet pass with self-attention replaced by identity gives $\boldsymbol{\epsilon}_{\text{weak}}$; the amplified prediction is $\boldsymbol{\epsilon}_{\text{stg}} = \boldsymbol{\epsilon}_{\text{full}} + s\,(\boldsymbol{\epsilon}_{\text{full}} - \boldsymbol{\epsilon}_{\text{weak}})$. TAG and STG act in parallel on the same IP2P prediction, $\tilde{\boldsymbol{\epsilon}}_{\text{tgt}} = \text{TAG}_{\eta}(\boldsymbol{\epsilon}_{\text{tgt}}) + s^{\text{eff}}_{\text{stg}}(\boldsymbol{\epsilon}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{weak}})$, with $s_{\text{stg}} = 3.0$ scaled by a **bump** over training progress (zero at the start, peak at 40%, zero again from 80%) and by $(1 - s)$ [J]. *Why bump:* late full-strength STG amplifies the *current* NeRF self-attention features, which on

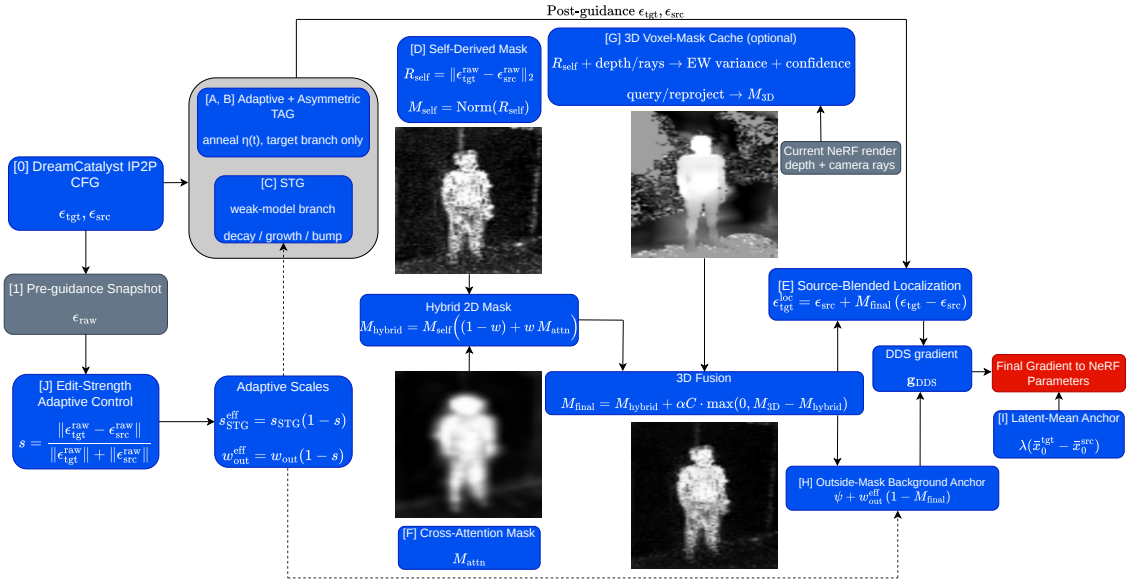


Figure 1: Method overview. DreamCatalyst's IP2P CFG is extended with modular guidance, localization, and preservation branches: STG, adaptive/asymmetric TAG, target- and source-instruction masks, cross-attention masking with an agreement floor, 3D voxel-mask aggregation, source-blended localization, outside-mask anchoring, latent-mean anchoring, and adaptive preservation controls. With every component disabled the pipeline is DreamCatalyst.

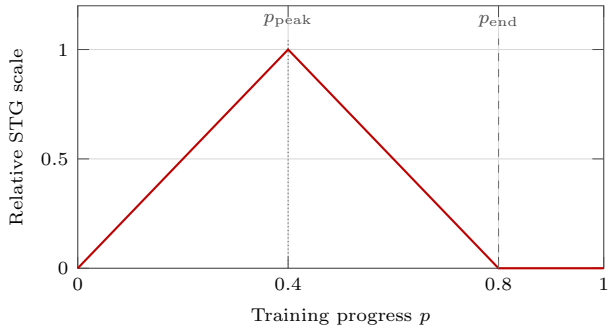


Figure 2: STG bump schedule, before the $(1-s)$ attenuation of [J]. STG helps during the mid-training commitment window and is off for the last 20% of training, avoiding per-view freezing on creative edits.

creative-transform scenes reflect view-dependent partial-state geometry (e.g. "different armor color per shoulder"). The bump gives mid-phase structural help and leaves the last fifth of training to TAG and localization. Conceptually aligned with DaCapo's temporal edit/preserve separation.

2.2 G2 — Localization

[D] Two instruction masks: one for the edit, one for preservation. With f the image-guided part of the IP2P CFG, the raw DDS delta splits exactly into what the instruction still *demands* of the current render and the *displacement* the render has already made:

$$\Delta\boldsymbol{\epsilon}^{\text{raw}} = \underbrace{s_{\text{txt}}\big(\boldsymbol{\epsilon}_{\text{txt}} - \boldsymbol{\epsilon}_{\text{img}}\big)(z_t)}_{\text{demand}} + \underbrace{f(z_t) - f(z_t^{\text{src}})}_{\text{displacement}}.$$

The thesis self-mask used $\|\Delta\boldsymbol{\epsilon}^{\text{raw}}\|$, so it rose wherever the render had drifted, wanted or not, and then admitted more edit force there. The final design separates the roles.

- **Edit mask M_{tgt}**: the demand alone, i.e. the per-pixel norm of $\boldsymbol{\epsilon}_{\text{txt}} - \boldsymbol{\epsilon}_{\text{img}}$ of the target pass (the instruction contrast of LatentEditor [16]), p_5–p_{95} normalized, smoothed by a per-view EMA with a memory of 5 visits of that view ($\beta = e^{-1/5}$), sharpened ($\gamma = 1.2$) and blurred. It follows the edit as it forms, so new structure can receive edit force. (The thesis EMA, $\beta = 1 - 1/(2N_{\text{cam}})$, advanced only on visits of its own view, so on the 315-view person scene each view's mask kept ≈99% of its first visit until the end of the run.)
- **Preservation mask M_{src}**: the same contrast on the *source* latent noised at $t = 500$, averaged over 4 noise draws (DiffEdit's recipe [12]), computed once per view and cached. The edit cannot change it, so a drifting region can never relax its own restoring force [H].

Both masks, like the edit strength [J], are read *before* TAG or STG modify the predictions, so they cannot inherit the amplification they are meant to restrain, consistent with DiffEdit's pre-guidance mask.

[E] Source-blended localization. Inspired by **LatentEditor** [16], we blend inside the DDS score rather than post-hoc:

$$\boldsymbol{\epsilon}^{\text{loc}}_{\text{tgt}} = \boldsymbol{\epsilon}_{\text{src}} + M \odot (\tilde{\boldsymbol{\epsilon}}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}), \tag{2}$$

so outside-mask regions fall back to the source branch before the DDS difference is computed. The blend acts on the guided prediction $\tilde{\boldsymbol{\epsilon}}_{\text{tgt}}$, so TAG and STG are localized together with the edit, and it masks only the DDS term:



Rendered Self-mask CA mask M_{attn} Fused mask

Figure 3: Mask hierarchy, from a run of the thesis design (DDS-delta self-mask, product fusion). The self-derived mask gives fine-grained localization and the cross-attention mask a semantic prior. The final design builds the self-mask from the instruction contrast [D] and bounds the attention veto with the source-instruction floor [F].

DreamCatalyst's preservation term stays unmasked and keeps pulling toward the source outside the mask. FoI [17] and ZONE [18] support the broader idea of moving localization *inside* the editing process.

[F] Cross-attention prior with an agreement floor. Following **Prompt-to-Prompt** [14], **DAAM** [15], and **LEDITS++** [13], we aggregate target-token cross-attention from UNet up-blocks $[1, 2]$ into M_{attn}. Target tokens are auto-derived from the prompts (target-only words minus stopwords and source words), with no manual per-scene keyword overrides. The product $M_{\text{tgt}} M_{\text{attn}}$ can only subtract support. At full strength from the first step it keeps the edit off the background, but it also vetoes parts of the requested edit that the token does not attend to: with the product alone, a remnant of the source hand remained beside the stormtrooper's glove. The thesis ramped the CA weight in late, which avoids that veto early but leaves the early, high-noise steps unlocalized. The final design keeps CA at full strength and bounds what it may veto:

$$M = \max\big(M_{\text{tgt}}\, M_{\text{attn}},\ \min(M_{\text{tgt}}, M_{\text{src}})\big). \tag{3}$$

Attention keeps the final say only where the two instruction masks disagree (structure beyond the source silhouette, spill the source does not support). M never exceeds M_{tgt}, the floor needs two measurements on different latents to agree, and it adds no hyperparameter.

2.3 G3 — 3D Cross-View Localization

[G] 3D voxel-mask cache for cross-view localization. The masks above are computed from one rendered view at a time. This is the core limitation behind the *clown* failure case: view 1 may strongly edit an arm while view 165 does not, and no per-view EMA can make those two rays talk to each other. To add a lightweight 3D consistency prior, we lift the current 2D edit mask into a non-parametric world-space voxel cache.

For a pixel (u, v) with rendered depth $D(u, v)$ and Nerfstudio world ray $(\mathbf{o}, \mathbf{d})$, we backproject

$$\mathbf{x}(u, v) = \mathbf{o} + D(u, v)\, \mathbf{d}. \tag{4}$$

The point is discretized into a voxel index $\mathbf{q}(\mathbf{x})$. If the ray terminates on a surface (rendered accumulation above 0.5), the voxel stores a running average of the *raw* target-instruction magnitude M_{raw} of [D], divided by its per-frame 95th percentile (the rank-normalized edit mask would destroy the absolute foreground/background contrast, and it already contains the attention restriction and the cache's own additions):

$$V_{\mathbf{q}} \leftarrow \beta_{\text{EMA}} V_{\mathbf{q}} + (1 - \beta_{\text{EMA}}) M_{\text{raw}}(u, v), \tag{5}$$

with $\beta_{\text{EMA}} = 1 - 1/(0.5\, N_{\text{cam}})$. A voxel's first observation is copied, so a long memory would leave the read-back "consensus" mostly the first observer's value throughout the fusion ramp. With the factor 0.5, a voxel seen by 40% of 315 cameras has about 9 effective observations behind its value when fusion starts and about 160 when it reaches full strength (about 2 and 4 with the thesis factor 2). At later views the same cache is queried by backprojecting their pixels into the same world grid, and the queried value is fused with the 2D mask M of [F] through a *positive-only* correction:

$$M_{\text{final}} = M + \alpha\, C\, G\, \max\big(M_{\text{3D}} - M,\, 0\big), \tag{6}$$

where α is a warmup-controlled blend, C is the per-voxel confidence (observation count, exponentially weighted cross-view variance, angular diversity and cached mass) and $G = \max(M_{\text{attn}}, M_{\text{tgt}})\, M_{\text{attn}}$ is a semantic gate. The factor M_{attn} matters: the cache is built from the same contrast as M_{tgt}, so M_{tgt} alone must not be able to authorize an addition. The cache adds edit support where the 3D consensus exceeds the 2D evidence, and never subtracts. A subtractive (bidirectional) variant, which also pulled the 2D mask down toward the 3D consensus, was evaluated and discarded: the averaged-across-views cache value systematically sits below the sharp 2D peaks, so the subtraction eroded genuine fine detail without improving consistency. The design is inspired by Semantic-NeRF-style rendering of scene-level fields [19] and by panoptic/semantic lifting of noisy 2D predictions into multi-view-consistent 3D representations [20]. InterGSEdit pursues a related cross-view-consistent editing goal in 3DGS via inter-view attention [21]; our cache is a lightweight, non-parametric NeRF-friendly alternative. In this thesis, the novelty is the *online supervision signal*: the 3D mask is built from the diffusion model's own instruction contrast during editing, not from precomputed segmentation labels.

2.4 G4 — Preservation Anchors

[H] Outside-mask background anchor. The DreamCatalyst preservation term $e^{\hat{f}}\psi\,(x_0^{\text{tgt}} - x_0^{\text{src}})$ is strengthened where the instruction does not apply: $\text{preserve}(x) = \psi + w_{\text{out}}\,(1 - s)\,(1 - M_{\text{src}}(x))$, with $w_{\text{out}} = 0.25$. It reads the fixed source mask, not the edit mask, so the brake is a fixed potential: a region that drifts toward the target cannot relax its own restoring force, and in Part 2 a cache

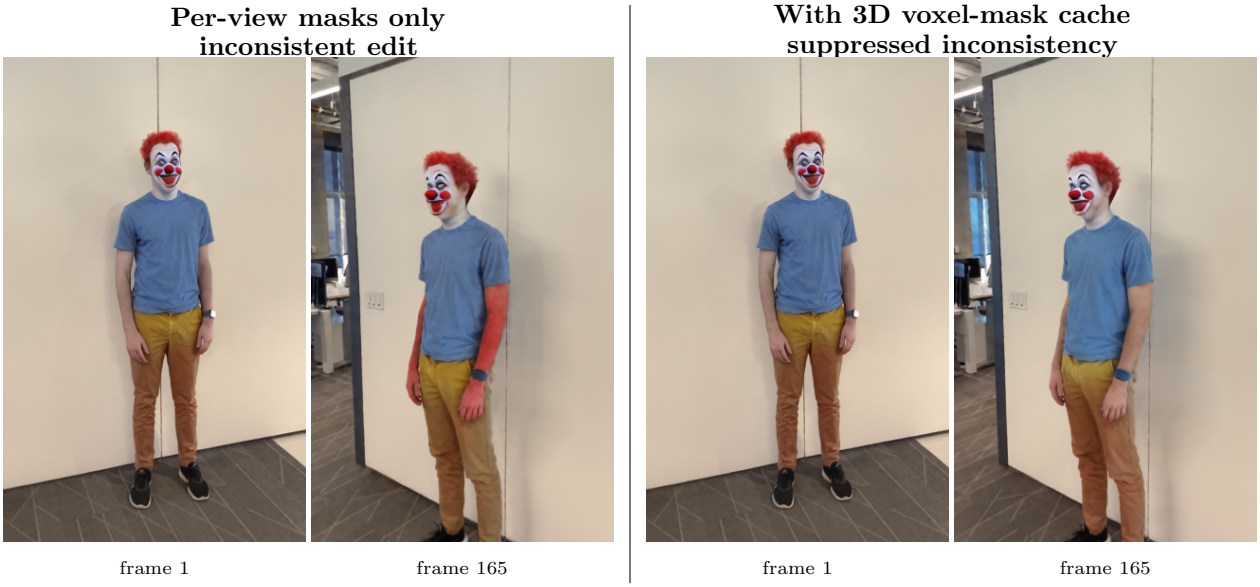


Figure 4: Qualitative motivation for the 3D voxel-mask cache. Without a 3D mask prior, the clown edit can become view-dependent: one training view accumulates arm/costume edits that another view does not. The voxel cache lifts high-confidence 2D mask evidence into world space and reprojects it to later views, suppressing this cross-view disagreement. A completed n=3 evaluation confirms that this benefit is qualitative: the cache makes residual failures view-coherent, while the aggregate image metrics stay within run-to-run noise.

false positive cannot gain edit force and lose preservation at the same time. Conceptually aligned with RoMaP's [10] spatial regularization.

[I] Latent-mean anchor. With TAG active we observed a systematic brightness and saturation drift of the per-channel means in VAE-latent space. We add a simple analytic bias to the final gradient:

$$\mathbf{g} \leftarrow \mathbf{g} + \lambda \left(\overline{x}_0^{\text{tgt}} - \overline{x}_0^{\text{src}}\right), \tag{7}$$

where $\overline{x}_0 = \text{mean}(x_0, \text{dim}{=}(H, W))$ is the per-channel spatial mean. This regularizes channel-mean drift directly in latent statistics. Spirit follows Piva [8] (explicit identity term) and SSD [9] (source-anchored stabilization). Empirical optimum on face: $\lambda = 0.005$.

2.5 G5 — Adaptive Mechanisms & Universal Configuration

Core idea. The same hyperparameter should not be optimal for identity-preserving scenes (face) and creative-transform scenes (stormtrooper). Instead of re-tuning per scene, we make preservation *self-adjust* based on a cheap proxy of scene type: the *edit strength* $s \in [0, 1]$ read directly from the raw noise predictions.

[J] Edit-strength-adaptive preservation. From the clean pre-guidance DDS snapshots $\boldsymbol{\epsilon}_{\text{tgt}}$ and $\boldsymbol{\epsilon}_{\text{src}}$ (captured before STG and TAG, like the masks of [D]), we define

$$s = \frac{\|\boldsymbol{\epsilon}_{\text{tgt}}^{\text{raw}} - \boldsymbol{\epsilon}_{\text{src}}^{\text{raw}}\|}{\|\boldsymbol{\epsilon}_{\text{tgt}}^{\text{raw}}\| + \|\boldsymbol{\epsilon}_{\text{src}}^{\text{raw}}\|} \in [0, 1],$$

which is bounded by the triangle inequality and time-invariant (numerator and denominator scale together with the timestep-dependent noise magnitude). Identity-preserving edits give $s\downarrow$ (target mostly agrees with source); structural edits give $s\uparrow$. We scale both the background anchor and the STG scale by $(1 - s)$:

$$w_{\text{out}}^{\text{eff}} = w_{\text{out}}\,(1 - s), \tag{8}$$

$$s_{\text{stg}}^{\text{eff}} = s_{\text{stg}}\,(1 - s). \tag{9}$$

A single $(w_{\text{out}}, s_{\text{stg}})$ pair serves every scene; bold edits loosen both automatically, subtle edits keep both near full strength.

Universal configuration. Combining the evaluated components (A–F and H–J) yields a single configuration tested across all 8 edits without per-scene tuning (see Sec. 3.1); the 3D voxel cache (G) is evaluated separately as an optional module (Sec. 3.3).

3 Experiments

Setup. NeRF (*nerfacto*) + DreamCatalyst-DDS with IP2P as the editor, 3000 edit iterations, fixed s_{txt}=7.5 and s_{img}=1.5 for fair comparison with the reference paper. Metrics: CLIP-direction ↑ (editability), CLIP-image ↑ and SSIM ↑ (content preservation), LPIPS ↓ (perceptual distance), multi-view cosine similarity ↑ (3D consistency), plus a custom *EditMaskVariance_3D* ↓ for Part 2. All numbers are means over n=3 seeds per configuration on a fixed reconstruction *checkpoint* per scene; run-to-run variation is ≈ 0.005–0.01 in CLIP-direction, which sets the bar for a real difference and underlies the **green** highlighting in Tables 1 and 2.

Scenes. Face/elf and Einstein are identity-preserving

face edits; clown and stormtrooper are full-body person edits (mixed makeup-plus-costume, and structural helmet growth); bear is a creative object-centric transformation; fangzhou is a localized structural addition (a knight's helmet); and Yuseung contributes two identity-with-costume edits, Batman and Joker.

3.1 Universal Configuration (Primary Result)

The universal configuration achieves (Table 1):

- **SSIM improved on all 8 edits** (mean 0.825→0.854), **LPIPS dropped on 7 of 8** (mean −9.2%; the only exception, Einstein, is within noise), and CLIP-I rose on 7 of 8. This is a clean, near-universal preservation gain.
- **Editability is preserved, not traded away**: mean $CLIP_{\text{dir}}$ 0.151 → 0.152, with the largest per-scene gains on harder edits (**Stormtrooper** +10.3%, Batman +5.8%, elf +4.1%).
- **Fine background detail** (wall/floor line patterns, background color fidelity, material textures) is consistently better preserved; this is most visible in the bear scene, where the baseline blurs background geometry.
- The few per-scene editability dips (bear, fangzhou, Einstein) stay within run-to-run noise: they reflect the honest editability/preservation balance of a *single, non-tuned* configuration rather than a regression.

Final configuration (fully general; no keywords or segmentation masks): adaptive asymmetric TAG ($\eta_{\text{tag}} = 1.2$); STG (scale 3.0, bump peaking at 40% and off from 80%, edit-strength-adaptive); edit mask M_{tgt} from the target-instruction contrast (per-view EMA, memory 5 visits, $\gamma = 1.2$); preservation mask M_{src} from the source-instruction contrast ($t = 500$, 4 noise draws, cached per view); cross-attention prior at full strength from the first step ($\gamma = 1.2$, up-blocks $[1, 2]$, auto-derived tokens) with the agreement floor; source-blended localization; an outside-mask anchor (0.25) on $1 - M_{\text{src}}$ with edit-strength-adaptive scaling; and a latent-mean anchor (0.005). Part 2 enables the voxel cache and changes nothing else.

Thesis-era configuration behind Table 1: DDS-delta self-mask with a camera-count EMA, CA weight on a power-0.75 schedule from 0 to 1, anchor 0.15 on the final mask, STG 3.5 (3.0 with the cache).

3.2 Scene-Type Analysis

Observation. Edit regime varies strongly by scene type. Edit-strength-adaptive preservation (J) converts this from a per-scene tuning problem into a property the method infers automatically from the raw DDS delta, without needing a mask.

Identity-preserving edits (Elf, Einstein): small edit strength ($s\downarrow$, target noise prediction stays close to source). Preservation must stay tight to maintain identity. Adaptive anchor keeps $w_{\text{out}}^{\text{eff}} \approx (1-s)\, w_{\text{out}}$ near w_{out}; STG stays near base scale.

Creative-transform edits (Stormtrooper, Panda): large edit strength ($s\uparrow$, target pulls the latent to a structurally different rendering). Preservation must loosen so the transformation can commit. Adaptive anchor automatically deflates $w_{\text{out}}^{\text{eff}}$; STG fades, avoiding its documented late-phase view-freezing failure mode. The bear scene additionally demonstrates the method's background geometry recovery: outside-mask anchoring explicitly restores floor line patterns and background color that unconstrained DDS erases as a side-effect of the edit.

Mixed edits (Clown, full-body costume plus face makeup): moderate edit strength. Preservation sits between the two extremes. The same hyperparameters serve all three regimes.

Quantitative note. The scalar s is logged throughout training as `dc_debug/edit_strength`, making the attenuation applied to STG and the outside-mask anchor directly inspectable per scene.

3.3 Part 2: Multi-View Consistency via the 3D Voxel Cache

Part 2. The universal configuration edits each view independently, so a region can be edited in one camera and preserved in another (the *clown*'s arm, Fig. 4). The 3D voxel cache (contribution G) is an *optional* module that lifts the per-view edit mask into a shared world grid and reprojects it to later views, so the edit *decision* becomes shared across cameras.

Evaluated separately (n=3, eight edits; Table 2) as an optional module on top of the universal configuration, the cache is **metric-neutral in aggregate**: every image metric, together with a custom mask-level 3D-consistency metric I introduced (*EditMaskVariance_3D*, which back-projects per-view edit magnitudes into a shared voxel grid and measures their cross-view variance), stays within run-to-run noise. Its contribution is **qualitative and structural**: residual failures become view-coherent, edited in all views or in none rather than flickering between them, which moves the stochasticity from the camera axis to the run axis. A controlled isolation on the *elf* scene (cache readback off / nearest-voxel / observed-weighted trilinear, everything else fixed) pinpoints the mechanism: the trilinear readback lifts rendered multi-view cosine similarity to 0.933 (vs. 0.924 nearest, 0.928 no cache) and lowers EMV_{3D} to ≈ 0.0225 at unchanged editability. The effect is small and near-noise, but it isolates what turns mask-level consistency into *rendered* consistency.

Why the readback matters. Both variants query the same cache; they differ only in how a world-space point is turned back into a scalar. Nearest lookup inherits the grid resolution directly, so a mask that is smooth in 3D returns to the 2D score with voxel-sized steps, and with speckle wherever adjacent voxels disagree or were never written. Plain trilinear interpolation removes the steps but introduces a second problem: it averages in voxels the cache has never observed, pulling the mask toward

Table 1: Universal configuration across 8 edits vs. DreamCatalyst (DC) baseline, same code + same prompts per scene, mean over n=3 runs. One fixed configuration (no per-scene tuning). Better value per cell in **green** when the gap exceeds run-to-run noise; near-ties are left unmarked.

Scene	CLIP_{dir} (DC → Ours)	CLIP-I (DC → Ours)	SSIM (DC → Ours)	LPIPS (DC → Ours)	MV Cos (DC → Ours)
Face / Tolkien elf	0.123 → **0.128**	0.592 → **0.601**	0.820 → **0.834**	0.256 → **0.231**	0.920 → **0.928**
Face / Einstein	0.081 → 0.078	0.557 → 0.559	0.794 → **0.806**	0.255 → 0.258	0.922 → 0.921
Person / Clown	0.138 → 0.143	0.773 → 0.772	0.924 → **0.935**	0.227 → **0.198**	0.911 → 0.910
Person / Stormtrooper	0.155 → **0.171**	0.650 → **0.682**	0.795 → **0.839**	0.331 → **0.295**	**0.926** → 0.901
Bear / Panda	**0.180** → 0.174	0.745 → **0.788**	0.693 → **0.768**	0.283 → **0.234**	**0.907** → 0.896
Fangzhou / Helmet	**0.206** → 0.182	0.727 → **0.764**	0.796 → **0.834**	0.311 → **0.282**	**0.951** → 0.943
Yuseung / Batman	0.156 → **0.165**	0.627 → **0.653**	0.869 → **0.889**	0.273 → **0.251**	0.930 → **0.935**
Yuseung / Joker	0.170 → 0.172	0.643 → **0.655**	0.908 → **0.923**	0.241 → **0.225**	0.932 → **0.938**
Mean	0.151 → 0.152	0.664 → **0.684**	0.825 → **0.854**	0.272 → **0.247**	0.925 → 0.922

Table 2: Part 2: universal config (2D, cache off) → with the 3D voxel cache (Vox.), per scene, mean over n=3 runs. The "2D" column is the universal configuration of Table 1; "Vox." adds the cache (with STG lowered to 3.0). EMV_{3D} is the custom mask-level 3D-consistency metric (lower is nominally better, though confounded by edit strength). Better value per cell in **green** only when the gap exceeds run-to-run noise: just 5 of 48 cells separate, which is the result itself, the cache is *metric-neutral* in aggregate, and its value is qualitative (Fig. 4).

Scene	CLIP_{dir} (2D→Vox.)	CLIP-I (2D→Vox.)	SSIM (2D→Vox.)	LPIPS (2D→Vox.)	MV Cos (2D→Vox.)	EMV_{3D} (2D→Vox.)
Face / elf	0.128 → 0.131	0.601 → 0.603	0.834 → 0.833	0.231 → 0.231	0.928 → 0.925	0.0243 → 0.0237
Face / Einstein	0.078 → 0.082	0.559 → 0.561	0.806 → 0.810	0.258 → **0.246**	**0.921** → 0.916	0.0097 → 0.0102
Person / Clown	0.143 → 0.144	0.772 → 0.781	0.935 → 0.935	0.198 → 0.198	0.910 → 0.907	0.0080 → 0.0079
Person / Stormtrooper	0.171 → **0.175**	0.682 → 0.676	0.839 → 0.838	0.295 → 0.293	0.901 → 0.899	0.0054 → 0.0059
Bear / Panda	**0.174** → 0.166	0.788 → 0.787	0.768 → 0.769	0.234 → 0.232	0.896 → 0.893	0.0338 → 0.0339
Fangzhou / Helmet	0.182 → 0.189	0.764 → 0.758	0.834 → 0.838	0.282 → **0.276**	0.943 → 0.943	0.0182 → 0.0188
Yuseung / Batman	0.165 → 0.162	0.653 → 0.651	0.889 → 0.888	0.251 → 0.252	0.935 → 0.934	0.0111 → 0.0104
Yuseung / Joker	0.172 → 0.172	0.655 → 0.662	0.923 → 0.922	0.225 → 0.225	0.938 → 0.935	0.0221 → 0.0207
Mean	0.152 → 0.153	0.684 → 0.685	0.854 → 0.854	0.247 → 0.244	0.922 → 0.919	0.0166 → 0.0164

zero exactly where view coverage is thinnest and the 3D prior is most needed. The implementation therefore masks each of the eight corner weights by an `observed` flag and renormalizes by the surviving weight,

$$M_{3D}(x) \;=\; \frac{\sum_{k=1}^{8} w_k\, o_k\, V_k}{\sum_{k=1}^{8} w_k\, o_k},$$

falling back to a neutral value only when no surrounding voxel has been seen at all. Unobserved corners are skipped rather than counted as zeros, which is what lets the smoothing act on the cached signal instead of eroding it. In the final design the cross-view variance that gates trust is read over the same eight corners (corners seen by at least two views, within-voxel variance only), so the agreement gate is as continuous as the value it gates.

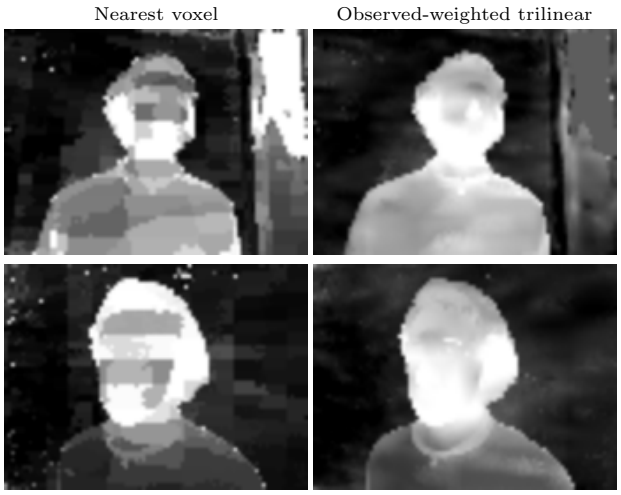


Figure 6: Voxel readback decides whether a 3D-consistent mask survives as a usable 2D mask. Cache-queried edit masks for two training views of the *elf* scene, with cache contents and every other setting held fixed; only the readback differs. Nearest-voxel lookup (left) returns the grid quantized: block edges across the subject and salt-and-pepper speckle on the background. Observed-weighted trilinear (right) returns a continuous mask with a smooth boundary. This is the difference the rendered multi-view cosine similarity reacts to (0.933 vs. 0.924).

The cache is non-parametric and builds its supervision online from the diffusion model's own DDS localization, never from precomputed segmentation. Its benefit is honestly per-scene rather than a headline number: on broad-recoloring edits it needs a variance-based agreement gate so that it does not also make a *coherent-but-unwanted* edit more consistent, and it inherits the quality of the per-view mask. I also report the mechanisms that did *not* work, a peak-hold variance latch and active suppression of contested regions, as documented negative results.

4 Methodological Findings

We highlight three methodological results that generalize beyond the specific experiments above:

(1) Mask construction must use a pre-guidance noise snapshot. Building M_{self} from post-guidance noise predictions makes the mask inherit the guidance stack's own distortions. TAG tangential amplification and STG structural extrapolation can both increase $\|\Delta\boldsymbol{\epsilon}\|$ in regions that are guidance artifacts rather than true edit targets. The pre-guidance noise snapshot from contribution D keeps localization independent of those later guidance modifications. This mirrors DiffEdit's [12] design.

(2) STG's effect is scene-type-dependent. On identity-preserving scenes, STG reinforces desirable structural details throughout training. On creative-transform scenes, late full-strength STG amplifies view-dependent partial-state features of the current NeRF, locking in inconsistencies. The **bump** schedule resolves this tension: STG helps during the mid-phase geometric commitment window but exits before the refinement stage.

(3) 3D consistency requires a 3D memory, not only a smoother 2D mask. Per-view EMA can reduce temporal flicker, but it cannot enforce agreement between two different camera views. The voxel cache is valuable because it stores edit evidence at an approximate 3D point and reuses it when another view observes that point. It does not solve semantic ambiguity by itself; it turns a good per-view localization signal into a more view-consistent one.

5 Metric Limitation

CLIP-direction alone is not sufficient for creative 3D edits. On the stormtrooper scene, one high-CLIPd run (0.211) produced a stormtrooper-patterned shirt rather than full armor. This shows that CLIP-direction can reward texture-level prompt evidence without measuring whether the intended 3D structure has been completed. For creative edits, I therefore report quantitative metrics together with qualitative renderings.

6 Limitations & Planned Extensions

The 3D voxel cache improves consistency, not the aggregate metrics. Across the completed n=3 evaluation the cache is metric-neutral: it makes the clown cross-view inconsistency view-coherent (Fig. 4) but does not move the averaged numbers, so its benefit is best read qualitatively and per scene. Its main limitation is that it inherits the quality of the per-view DDS mask: if the 2D mask is semantically wrong, the cache can make that wrong signal more consistent.

The STG schedule shape is fixed, not inferred. Edit-strength-adaptive scaling adjusts the STG *magnitude*, but the bump shape is used for every scene, although the thesis runs suggested that the best shape depends on the edit's trajectory. Making the shape itself driven by s is a small extension.

Shrinking-silhouette edits can leave a "ghost hand". When an edit makes a region *smaller* than the source (the *stormtrooper*'s gloved hand is smaller than the person's original hand), the part of the old silhouette no longer covered should be *erased*; instead the outside-mask preservation anchor pulls it back toward the source, leaving a translucent residue. By inspection this appears more in the proposed configuration than in the baseline, whose weaker localization edits the region more freely. It is structural to a localize-and-preserve design and motivates a future "erase-aware" anchor that releases preservation where the silhouette shrinks.

Evaluation extension: cross-scene human preference study, including the "CLIP-metric gaming" failure case, to quantify when quantitative and qualitative evaluation disagree.

Refinement stage (IV): not evaluated in this thesis, matching DreamCatalyst's own protocol. Contributions operate entirely in stage (III).

7 Technical Context

DreamCatalyst's DDS uses time-weighted $w_{\text{DDS}}(\hat{t}) = \delta + \gamma\hat{t}^{1/e}$ (δ=0.2, γ=0.8). The target branch uses IP2P's 3-way CFG:

$$\boldsymbol{\epsilon}_{\text{tgt}} = \boldsymbol{\epsilon}_{\varnothing} + s_{\text{txt}}(\boldsymbol{\epsilon}_{\text{txt}} - \boldsymbol{\epsilon}_{\text{img}}) + s_{\text{img}}(\boldsymbol{\epsilon}_{\text{img}} - \boldsymbol{\epsilon}_{\varnothing}).$$

Our contributions operate between CFG and the DDS gradient, but mask construction and mask application occur at different points. Before any amplification, M_{tgt} and M_{attn} are read from the target pass, M_{src} from the cached source passes, the edit strength s from the CFG-combined predictions, and the edit mask M (plus the voxel-cache addition M_{final}, from the current NeRF depth and rays) is built from them. STG and TAG then modify the target prediction. Finally, source-blended localization applies M_{final} to the guided delta $\tilde{\boldsymbol{\epsilon}}_{\text{tgt}} - \boldsymbol{\epsilon}_{\text{src}}$, the preservation term is weighted by $1 - M_{\text{src}}$, and the latent-mean anchor

is added. Each component can be switched off; with all of them off the pipeline is exactly DreamCatalyst.

8 Author Profile

I am an aspiring AI researcher who recently completed my B.Sc. in Control and Automation Engineering at UFSC (Brazil), graduating in 4.5 years, one semester ahead of the standard 5-year program (diploma pending). My interests span **generative modeling and its optimization (diffusion guidance, score distillation, 3D scene representations)**, **multimodal learning**, and **medical AI**, and I am strongly motivated to contribute to impactful research. Selected graduate-level coursework: Machine Learning (10/10), Computer Vision (10/10), Generative AI (9.5/10), Artificial Intelligence (9.5/10), and Data-Driven Algorithms (9/10). I am an AI intern at Upstart13 (US-based, remote) and previously interned at RT Medical Systems (radiotherapy/software company), where I took a medical-image segmentation pipeline from data extraction to deployment (nnU-Net to OHIF, write-up).

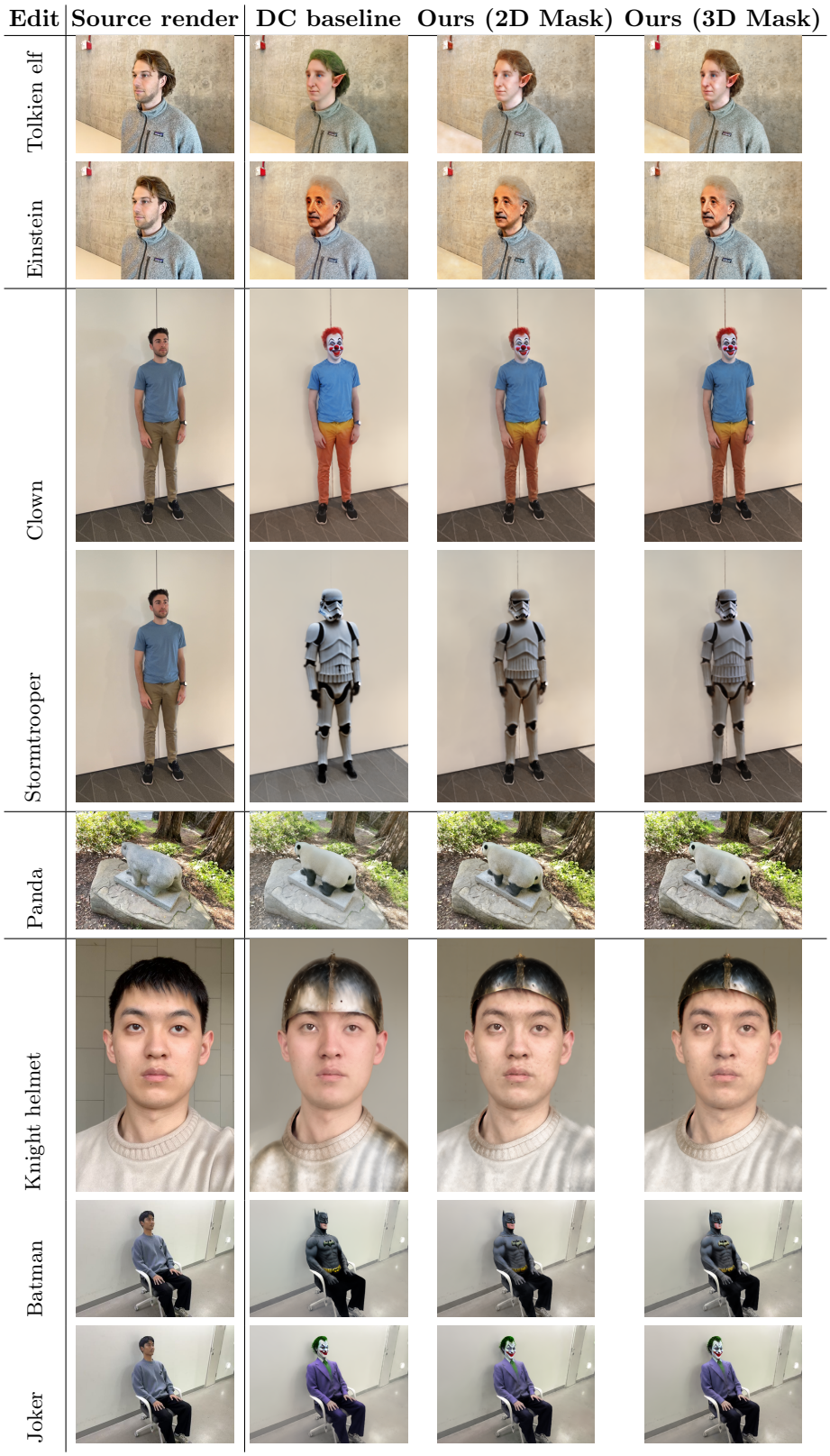


Figure 5: Qualitative results across all eight edits. Columns: source render, the DreamCatalyst baseline, our single universal configuration (Part 1, Sec. 3.1; no per-scene tuning), and the same configuration with the 3D voxel cache enabled (Part 2, Sec. 3.3). The universal configuration commits the edit while preserving background line patterns, color fidelity, and material textures more reliably (most visible in the bear scene). Enabling the voxel cache gives visually comparable single frames, consistent with its metric-neutrality (Table 2); its benefit is *cross-view* consistency (Fig. 4), not single-frame appearance. Full metrics: Tables 1 (Part 1) and 2 (Part 2).

References

[1] J. Kim, S. Lee, J. Shin, J. Choi, H. Shim, "DreamCatalyst: Fast and High-Quality 3D Editing via Controlling Editability and Identity Preservation," *ICLR*, 2025.

[2] A. Hertz et al., "Delta Denoising Score," *ICCV*, 2023.

[3] T. Brooks, A. Holynski, A. A. Efros, "InstructPix2Pix: Learning to Follow Image Editing Instructions," *CVPR*, 2023.

[4] H. Cho, D. Ahn, S. Hong, J. E. Kim, S. Kim, K. H. Jin, "TAG: Tangential Amplifying Guidance for Hallucination-Resistant Sampling," *ICML*, 2026.

[5] J. Hyung et al., "Spatiotemporal Skip Guidance for Enhanced Video Diffusion Sampling," *CVPR*, 2025.

[6] J. Koo, C. Park, M. Sung, "Posterior Distillation Sampling," KAIST, *CVPR*, 2024.

[7] Huang et al., "DaCapo: Score Distillation as Stacked Bridge for Fast and High-quality 3D Editing," *CVPR*, 2025.

[8] "Preserving Identity with Variational Score for General-purpose 3D Editing (Piva)," 2024.

[9] "Stable Score Distillation (SSD)," 2025.

[10] H. Kim, J. H. Jang, S. Y. Chun, "RoMaP: Robust 3D-Masked Part-level Editing in 3D Gaussian Splatting with Regularized SDS," SNU, 2025.

[11] Miao et al., "Rethinking Score Distilling Sampling for 3D Editing and Generation (UDS)," *ICML*, 2025.

[12] G. Couairon, J. Verbeek, H. Schwenk, M. Cord, "DiffEdit: Diffusion-based Semantic Image Editing with Mask Guidance," *ICLR*, 2023.

[13] M. Brack et al., "LEDITS++: Limitless Image Editing using Text-to-Image Models," *CVPR*, 2024.

[14] A. Hertz et al., "Prompt-to-Prompt Image Editing with Cross-Attention Control," *ICLR*, 2023.

[15] R. Tang et al., "What the DAAM: Interpreting Stable Diffusion Using Cross Attention," *ACL*, 2023.

[16] U. Khalid et al., "LatentEditor: Text Driven Local Editing of 3D Scenes," *ECCV*, 2024.

[17] Y. Guo, X. Lu, H. Lin, "Focus on Your Instruction: Fine-grained and Multi-instruction Image Editing by Attention Modulation," *CVPR*, 2024.

[18] J. Li, P. Liang, Y. Liu, S. Xie, "ZONE: Zero-Shot Instruction-Guided Local Editing," *CVPR*, 2024.

[19] S. Zhi, T. Laidlow, S. Leutenegger, A. Davison, "In-Place Scene Labelling and Understanding with Implicit Scene Representation," *ICCV*, 2021.

[20] Y. Siddiqui et al., "Panoptic Lifting for 3D Scene Understanding with Neural Fields," *CVPR*, 2023.

[21] M. Wen et al., "InterGSEdit: Inter-View Consistent 3D Gaussian Splatting Editing," 2025.